

1.1.2 Defining GIS

Q. 1.1.3 Define GIS with some fundamental observation. (Ref. sec. 1.1.2)

(5 Marks)

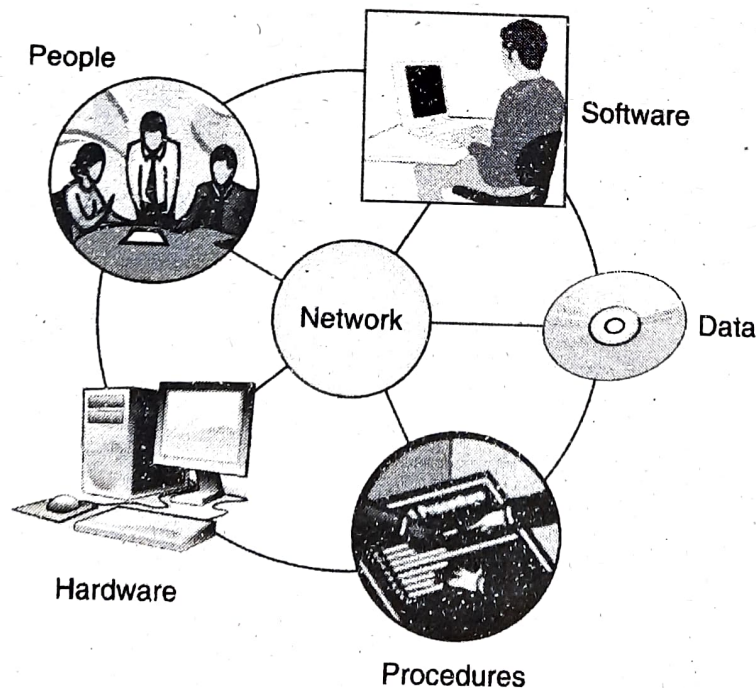
GIS is multidisciplinary (Engineering, Computer Science, Statistics, Mathematics, Geography, Psychology, Philosophy) research that addresses the nature of geographic information and the application of geospatial technologies to basic scientific questions.

Examples

1. Map generalization
2. Ontology

**Syllabus Topic : GISystems****1.1.3 GISystems****Q. 1.1.4 Explain GISystems (Ref. Sec. 1.1.3)****(5 Marks)**

- It is called as Geospatial/Geographic Information System. Geographic Information System (GISystem) is the most used concept of GIS. Many definitions of GISystem have been suggested over the years.
- In general, one could derive the practical definition of GISystem as a computerized system designed to dealing with the collection, storage, manipulation, analysis, visualization and displaying geographic information. GISystem is a tool to perform the spatial analysis which will put insight to the activities and phenomena carrying out every day.
- **GISystem** : Talks about GIS as a tool or system for solving geospatial related problems.
- Its components include : *Components of GIS System*
 1. Hardware
 2. Software
 3. Data/Information
 4. Users/People
 5. Procedures/Methods and
 6. Network

**Fig. 1.1.2 : The major components of GISystem**



Syllabus Topic : GIScience and GIApplications

1.1.4 GIScience and GIApplications

Q. 1.1.5 Explain GIScience and GIApplications.

(Ref. Sec. 1.1.4)

(5 Marks)

- Geographic Information Science (GIScience) is advocated to address a set of intellectual and scientific questions which go well beyond the technical capabilities of GISystem. The concept was first advocated by Michael Goodchild.
- GIScience talks about GIS as a scientific discipline of study in the academia. This is the science behind the technology aimed at enhancing knowledge of Geospatial concepts and their computational implementations. The major contributing disciplines are :

GIS application

1. Computer science
2. Mathematics/Statistics
3. Geomatics (Land Surveying, Photogrammetry, Remote Sensing, Geodesy, 'GPS, Drone mapping)
4. Geography and
5. Cartography

- GIS modeling GIS modeling: the use of GIS in the process of building the use of GIS in the process of building models with spatial data.
- Basic requirement in modelling : Knowledge of the systems be modeled be modeled.

Syllabus Topic : Maps

1.2.2 Maps

(5 Marks)

Q. 1.2.2 Explain maps with respect to GIS. (Ref. Sec. 1.2.2)

- A map is a visual representation of an area (can be for any space not just geographical)
More importantly, Maps helps us make sense of the world.

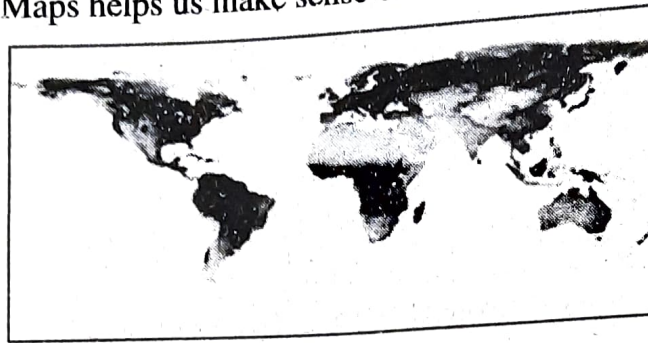
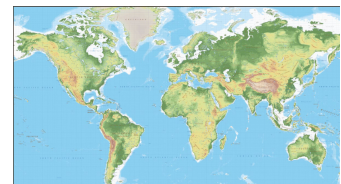


Fig. 1.2.1 : World Map of Vegetation on Earth

Types of maps

In practice we normally think of two types of map :

1. Topographic map
2. Thematic map



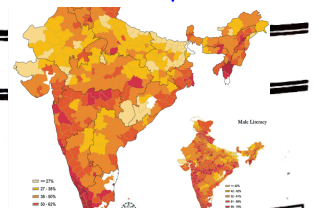
- **1. Topographic map** : A reference tool, showing the outlines of selected natural and man-made features of the Earth often acts as a frame for other information

“Topography” refers to the shape of the surface, represented by contours and/or shading, but topographic maps also show roads and other prominent features

- **2. Thematic map** : A tool to communicate geographical concepts such as the distribution of population densities, climate, movement of goods, land use etc. ↗

Syllabus Topic : Databases

1.2.3 Databases



Q. 1.2.3 Explain databases with respect to GIS. (Ref. Sec. 1.2.3)

(5 Marks)

Database with respect to GIS

- (A spatial database system may be defined as a database system that offers spatial data types in its data model and query language,) and supports spatial data types in its implementation, providing at least spatial indexing and spatial join methods.
- Spatial database systems offer the underlying database technology for geographic information systems and other applications. We survey data modeling, querying, data structures and algorithms, and system architecture for such systems. The emphasis is on describing known technology in a coherent manner, rather than listing open problems.
- (A spatial database therefore has the following characteristics :
 1. A spatial database system is a database system. Find all restaurants within 5 kilometers."
 2. It offers Spatial Data Types (SDTs) in its data model and query language. "What are the intersecting boundaries of two areas?"
 3. It supports spatial data types in its implementation, providing at least spatial indexing and efficient algorithms for spatial join. Merging data based on spatial relationships.

Syllabus Topic : Spatial Databases and Spatial Analysis

1.2.4 Spatial Databases and Spatial Analysis

Database with respect to GIS

Q. 1.2.4 Explain spatial databases and spatial analysis. (Ref. Sec. 1.2.4) (5 Marks)

- (A spatial database includes collections of information about the spatial location, relationship and shape of topological geographic features and the data in the form of attributes.) The design of the spatial database is the formal process of analyzing facts about the real world into a structured model.
- (Database design is characterized by the following phases :
 1. Requirement Analysis.
 2. Logical Design.
 3. Physical Design.
- In other words, you basically need a plan, a design layout and then the data to complete the process.)
- Having a solid well designed spatial database is the key to performing good Spatial Analysis. The database can be complex and designed with expensive sophisticated software or can be merely a simple well organized collection of data that can be utilized in a geographic form.



- Three main categories of spatial modeling functions that can be applied to geographic features within a GIS are :
 1. Geometric models, such as calculating the Euclidean distance between features, generating buffers, calculating areas and perimeters, and so on...
 2. Coincidence models, such as topological overlay
 3. Adjacency models (path finding, redistricting, and allocation).
- All three model categories support operations on spatial data such as points, lines, polygons, tins, and grids. Functions are organized in a sequence of steps to derive the desired information for analysis.
- Almost all entities of geographic reality have at least a 3-dimensional spatial character, but not all dimensions may be needed.

Steps in database design

1. Conceptual

- (a) Software and hardware independent.
- (b) Describes and defines included entities.
- (c) Identifies how entities will be represented in the database.
- (d) Selection of spatial objects - points, lines, areas, raster cells.
- (e) Requires decisions about how real-world dimensionality and relationships will be represented.
- (f) These can be based on the processing that will be done on these objects.

2. Logical

- (a) Software specific but hardware independent.
- (b) Sets out the logical structure of the database elements, determined by the data base management system used by the software.

3. Physical

- (a) Both hardware and software specific.
- (b) Requires consideration of how files will be structured for access from the disk.

Characteristics of a Good Database Design

In order that the GIS database provides the best service it should be :

- **Contemporaneous** : The data should be updated regularly so as to yield information that pertains to the same time-frame for all its measured variables.



- Flexible and extensible so that additional datasets may be added as necessary for the intended applications.
- **Positionally accurate** : (If for example the boundary between the residential and agricultural land has changed, this may be incorporated with ease.)
- Exactly compatible with other information that may be overlain with it.
- Internally accurate, portraying the nature of phenomena without error - requires clear definitions.
- Readily updated on a regular schedule.
- Accessible to whoever needs it.

1.3 Geographic Information and Spatial Database Models and Representations of the Real World

Syllabus Topic : Geographic Phenomena : Defining Geographic Phenomena

1.3.1 Geographic Phenomena

Definition

Q. 1.3.1 Define geographic phenomena. (Ref. Sec. 1.3.1)

(5 Marks)

(We might define a geographic phenomenon as a manifestation of an entity or process of interest that :)

- ✓ Can be named or described.
- ✓ Can be georeferenced.
- ✓ Can be assigned a time (interval) at which it is/was present.
- A GIS operates under the assumption that the spatial phenomena involved occur in a two- or three-dimensional Euclidean space. Euclidean space can be informally defined as a model of space in which locations are represented by coordinates (x, y) in 2D and (x, y, z) in 3D space-and distance and direction can defined with geometric formulas.
- In 2D, this is known as the Euclidean plane. To represent relevant aspects of real world phenomena inside a GIS, we first need to define what it is we are referring to.
- Relevance of phenomena for the use of a GIS depends entirely on the objectives of the study areas.
- ✓ (For instance, in water management, relevant objects can be river basins, agro-ecological units, measurements of actual evapotranspiration, meteorological data, groundwater



levels, irrigation levels, water budgets and measurements of total water use. All of these can be named or described, georeferenced and provided with a time interval at which each exists. In multipurpose cadastral administration, the objects of study are different :

- Houses
- Land parcels
- Streets of various types
- Land use forms
- Sewage canals and other
- Forms of urban infrastructure may all play a role.

Syllabus Topic : Types of Geographic Phenomena

1.3.2 Types of Geographic Phenomena

Q. 1.3.2 Explain types of geographic phenomena. (Ref. Sec. 1.3.2)

(5 Marks)

Three Types of geographical phenomena

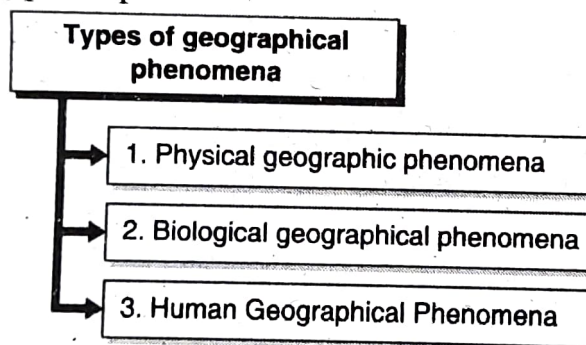


Fig. 1.3.1 : Types of geographic phenomena

→ 1. Physical Geographic Phenomena

- Physical geographic phenomena refer to those that are generated without involving any living organism. These drastic changes usually occur as a result of climatic, physical or chemical elements, among others, naturally generated.
- Within physical geographic changes can be hurricanes, cyclones, torrential rains and earthquakes, among others. The physical geographic changes that are generated are able to transform the landscape and generate a new reality.
- Examples of physical geographic changes can be :
 - (i) Rivers overflow (ii) Volcano eruption

→ (i) Rivers overflow

- A river can overflow as a result of different natural causes. Some of the possible causes may be the following :
 - o As a result of heavy rains and sustained in a short time.
 - o Consistent rainfall for a long time.
 - o Obstruction of channels by landslides.
 - o Ascent of sea level.
- When a river overflows, it can generate lasting changes in the landscape. It is possible for the river to permanently expand its channel, to inundate surrounding plant species, and if there are human communities nearby, it is possible that it may collapse with houses, buildings, roads and other constructions.

→ (ii) Volcano eruption

- The eruption of a volcano is generated by the movement tectonic plates. Or by the accumulation of pressure of the magma (molten rock that exists in the depths).
- In either case, the eruption of a volcano is considered a physical geographic phenomenon because it occurs without the intervention of living individuals.
- When a volcano erupts, it can have definite consequences on its environment, in which they stand out :
 - o The destruction of fauna as a result of the lava
 - o Generation of forest fires

→ 2. Biological geographic phenomena

(Biological geographic phenomena are those that are generated by living beings, excluding humans.) (Within this classification are the geographic variations produced by plants, animals, insects and microorganisms.) Some examples of biological geographic changes can be:

(i) Plague deforestation

(ii) Species extinction

eg.

→ (i) Plague deforestation

(The appearance of pests can destroy large areas of flora.) Pests especially affect plants located on land with few nutrients, which can lead to deforestation of entire regions and altering the environment completely. Pests can also greatly reduce the amount of animal organisms in an area.

→ (ii) Species extinction

- It is possible that a species disappears by natural causes, without the intervention of the human being. For example as a result of physical phenomena, such as droughts, fires or floods, among others. *Ex. drought, flood, etc.*
- The extinction of a species can permanently alter the entire ecosystem of which it forms part. The cycles of nature can vary, as can the structure of food chains.

→ 3. Human Geographical Phenomena

(These phenomena are the most obvious and, in many cases, invasive that can be found on the planet. Human geographic phenomena are caused exclusively by the action of man in his environment.) Like physical and biological phenomena, human geographic phenomena alter the environment in a lasting way. As a result of these transformations, positive and, in many cases, negative consequences can be generated.

Some examples of human geographic phenomena :

- Construction of roads
- Construction of dams

→ (i) Construction of roads

- As a result of the need to expand its communication channels, human beings have transformed their environment. This has involved the construction of roads and roads that intervene openly in the environment.
- The construction of this type of structures has been beneficial for the development of the human race, allowing to extend the interaction between the men and to generate a more effective communication.
- However, in some cases the intervention has been detrimental to nature, because some ecosystems have been affected.
- As a consequence of this type of constructions, entire species of flora and fauna can disappear, or deviations of water courses can be generated, among other manifestations.

→ (ii) Construction of dams

- Hydraulic dams are structures, made of walls and containment elements, whose main function is to store or divert water from a river to serve different purposes.
- Among the functions of a water dam is the regulation of water supply in a particular region, the storage of water for irrigation or energy production.

When constructing a dam, the human being intervenes to a great extent in the nature. These constructions generate positive consequences for human life, such as renewable energy production, flood control in certain areas and facilitating access to water for human consumption.

On the other hand, the construction of dams is considered a geographical phenomenon because it transforms the environment permanently :

- Generates stagnant water, which can bring disease.
- Block the passage to different marine species, affecting migratory movements.
- Promotes the extinction of whole colonies of organisms, which make life in the rivers.

Syllabus Topic : Geographic Fields

1.3.3 Geographic Fields

0.1.3.3 Explain the concept of geographic fields. (Ref. Sec. 1.3.3)

(5 Marks)

- The basic concept of a field came from physics. Geographers have developed independent theories, data models, and analytical methods. One reason for this apparent disconnect is that "geographic fields" tend to have a different fundamental nature than physical fields; that is, they have patterns similar to gravity and magnetism, but are in reality very different.
- Common types of geographic fields include :

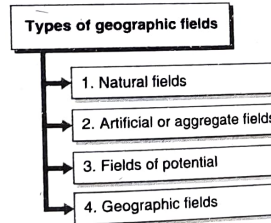


Fig. 1.3.2 : Types of geographical fields

→ 1. Natural fields

(Properties of matter that are formed at scales below that of human perception, such as temperature or soil moisture.)

**→ 2. Artificial or aggregate fields**

(Statistically constructed properties of aggregate groups of individuals, such as population density.)

→ 3. Fields of potential

(Which measure conceptual, non-material quantities (and are thus most closely related to the fields of physics), such as the probability that a person at any given location will prefer to use a particular facility (e.g. a grocery store).)

→ 4. Geographic fields

(Can exist over a temporal domain as well as space.) For example, temperature varies over time as well as location in space.) In fact, many of the methods used in time geography and similar spatiotemporal models treat the location of an individual as a function or field over time.

Syllabus Topic : Geographic Objects

1.3.4 Geographic Objects

Q. 1.3.4 Explain the geographic objects term with respect to GIS.

(Ref. Sec. 1.3.4)

(5 Marks)

- G of GIS can be interpreted as Geographic Object or Spatial Object or Spatial feature or Geographic Individuals. At the planet scale, our world is only one object with the defined boundary. At the global scale, you can view our planet as water and land mass. They have their own defined boundaries. In our planet, there is more water mass than land mass.
- (It is interesting why our planet is called Earth rather than Water. Anyway, at the more refined scale, electric poles, rivers, roads, tramlines, water pipes, fire hydrants, lakes, vineyards, agricultural land and the forest patches are examples of geographic objects.)
- Most of the tangible geographic objects have the defined acceptable boundary and properties such as name, types and status etc. These are called discrete geographic objects. These are also called discrete objects.
For example, Road is a object then it is characterized by :
 - **Location** : Where does it begin and end.
 - **Shape** : How many lines does it have?
 - **Size** : How far one can travel on it.
 - **Orientation** : In which direction one can travel ?

1.4 Computer Representations of Geographic Information

Syllabus Topic : Regular Tessellations

1.4.1 Regular Tessellations

Q. 1.4.1 Write a short note on regular tessellations.

(Ref. Sec. 1.4.1)

(5 Marks)

- ✓ A regular tessellation is a highly symmetric tessellation made up of congruent regular polygons. Only three regular tessellations exist: those made up of equilateral triangles, squares, or hexagons. honeycomb
- ✓ A tessellation or tiling of the plane is a collection of plane figures that fills the plane with no overlaps and no gaps. One may also speak of tessellations of the parts of the plane or of other surfaces. The word “tessella” means “small square”.



➤ Tessellations and color

- When discussing a tiling that is displayed in colors, to avoid ambiguity one needs to specify whether the colors are part of the tiling or just part of its illustration. See also color in symmetry.

The four color theorem states that for every tessellation of a normal Euclidean plane, with a set of four available colors, each tile can be colored in one color such that no tiles of equal color meet at a curve of positive length. Note that the coloring guaranteed by the four-color theorem will not in general respect the symmetries of the tessellation. To produce a coloring which does, as many as seven colors may be needed, as in the picture at right.

➤ Tessellations with quadrilaterals

- Copies of an arbitrary quadrilateral can form a tessellation with 2-fold rotational centers at the midpoints of all sides, and translational symmetry with as minimal set of translation vectors a pair according to the diagonals of the quadrilateral, or equivalently, one of these and the sum or difference of the two.
- For an asymmetric quadrilateral this tiling belongs to wallpaper group p2. As fundamental domain we have the quadrilateral. Equivalently, we can construct a parallelogram subtended by a minimal set of translation vectors, starting from a rotational center. We can divide this by one diagonal, and take one half (a triangle) as fundamental domain. Such a triangle has the same area as the quadrilateral and can be constructed from it by cutting and pasting.

Syllabus Topic : Irregular Tessellations

1.4.2 Irregular Tessellations

Q. 1.4.2 Write a short note on irregular tessellations (Ref. Sec. 1.4.2)

(5 Marks)

An irregular tessellation is a tessellation made of irregular shapes. Irregular tessellations can be created by a rule, or they can be created creatively. The more interesting tessellations are, of course, the creative ones, where people try to imagine what shapes might work well together, they try to create them.

As far as a rule method goes, you might be :

1. Start with hexagons, triangles, squares, rectangles, or other regular shapes that can be tessellated as well as assembled with many other identical shapes to cover a plane with no gaps or overlaps.
2. Start adding or subtracting other shapes to one edge of that shape while at the same time, subtracting or adding that same shape to another edge of that shape. For this step, it is probably easier to do the opposite on the opposite edge, and then as you get good at doing it this way, you can experiment to see if you can come up with a way of doing it in a different way. You could even add or subtract to the same edge.

Syllabus Topic : Vector Representations

1.4.3 Vector Representations

Q. 1.4.3 Explain in detail vector representations. (Ref. Sec. 1.4.3)

(5 Marks)

- (Vector is a data structure, used to store spatial data.) (Vector data is comprised of lines or arcs, defined by beginning and end points, which meet at nodes.) The locations of these nodes and the topological structure are usually stored explicitly.
- (Features are defined by their boundaries only and curved lines are represented as a series of connecting arcs.) Vector storage involves the storage of explicit topology, which raises overheads, however it only stores those points which define a feature and all space outside these features is 'non-existent'.
- A vector based GIS is defined by the vectorial representation of its geographic data. According with the characteristics of this data model, geographic objects are explicitly represented and, within the spatial characteristics, the thematic aspects are associated.
- (There are different ways of organising this double data base (spatial and thematic). Usually, vectorial systems are composed of two components: the one that manages spatial data and the one that manages thematic data. This is the named hybrid organisation system, as it links a relational data base for the attributes with a topological one for the spatial data.
- A key element in these kind of systems is the identifier of every object. This identifier is unique and different for each object and allows the system to connect both data bases.

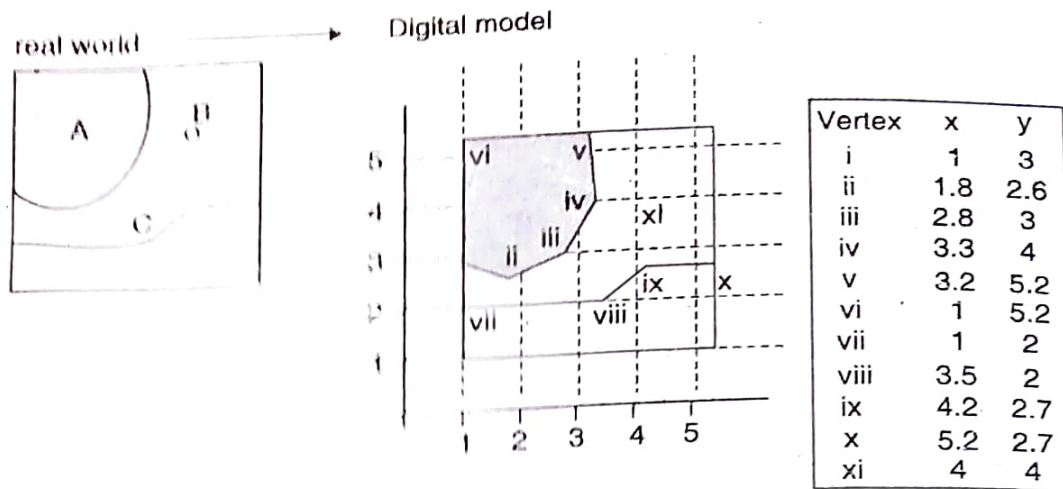


Fig. 1.4.1 : Vector representation

Vector representation of data

- In the vector based model (above diagram), geospatial data is represented in the form of co-ordinates. In vector data, the basic units of spatial information are points, lines (arcs) and polygons. Each of these units is composed simply as a series of one or more co-ordinate points.
- For example, a line is a collection of related points, and a polygon is a collection of related lines.

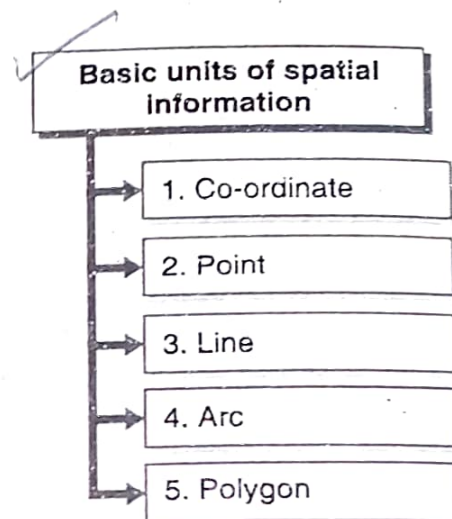


Fig. 1.4.2 : Basic units of spatial information

→ 1. Co-ordinate

Pairs of numbers expressing horizontal distances along orthogonal axes, or triplets of numbers measuring horizontal and vertical distances, or n-numbers along n-axes expressing a precise location in n-dimensional space. Co-ordinates generally represent locations on the earth's surface relative to other locations.

→ 2. Point

A zero-dimensional abstraction of an object represented by a single X,Y co-ordinate. A point normally represents a geographic feature too small to be displayed as a line or area;



for example, the location of a building location on a small-scale map, or the location of a service covers on a medium scale map.

→ 3. Line

A set of ordered co-ordinates that represent the shape of geographic features too narrow to be displayed as an area at the given scale (contours, street centerlines, or streams), or linear features with no area (county boundary lines). A line is synonymous with an arc.

→ 4. Arc

An ARC/INFO term that is used synonymously with line.

→ 5. Polygon

A feature used to represent areas. A polygon is defined by the lines that make up its boundary and a point inside its boundary for identification. Polygons have attributes that describe the geographic feature they represent.

Syllabus Topic : Topology and Spatial Relationships

1.5 ✓ Topology and Spatial Relationships

Q. 1.5.1 Explain topology and spatial relationships in detail. (Ref. Sec. 1.5) **(5 Marks)**

- Topology has long been a key GIS requirement for data management and integrity. (In general, a topological data model manages spatial relationships by representing spatial objects (point, line, and area features) as an underlying graph of topological primitives : nodes, faces, and edges.)
- These primitives, together with their relationships to one another and to the features whose boundaries they represent, are defined by representing the feature geometries in a planar graph of topological elements.

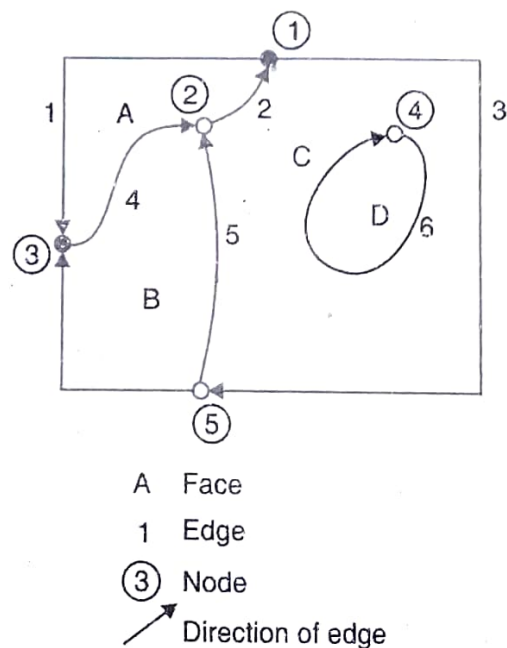


Fig. 1.5.1 : Topological elements and relationships



- Topology is fundamentally used to ensure data quality of the spatial relationships and to aid in data compilation. Topology is also used for analyzing spatial relationships in many situations, such as dissolving the boundaries between adjacent polygons with the same attribute values or traversing a network of the elements in a topology graph.
- Topology deals with spatial and structural properties of geometric objects, independent of their extension, type, or geometric form. Among the types of topological properties of objects there are the numbers of dimensions an object has or the relationships that exist between objects.

Syllabus Topic : Scale and Resolution

1.6 Scale and Resolution

(5 Marks)

Q. 1.6.1 Write a short note on scale and resolution. (Ref. Sec. 1.6)

- GIS data is stored in a very different way than paper map data, the relationships between map scale, data accuracy, resolution, and density are different between GIS and paper maps.)
- (A map in a GIS can be shrunk or enlarged at will on the screen or on paper. You can zoom in until the screen displays a square meter or less, or zoom out until the screen displays the entire earth.) (This means that geographic data in a GIS doesn't really have a 'map scale'.)

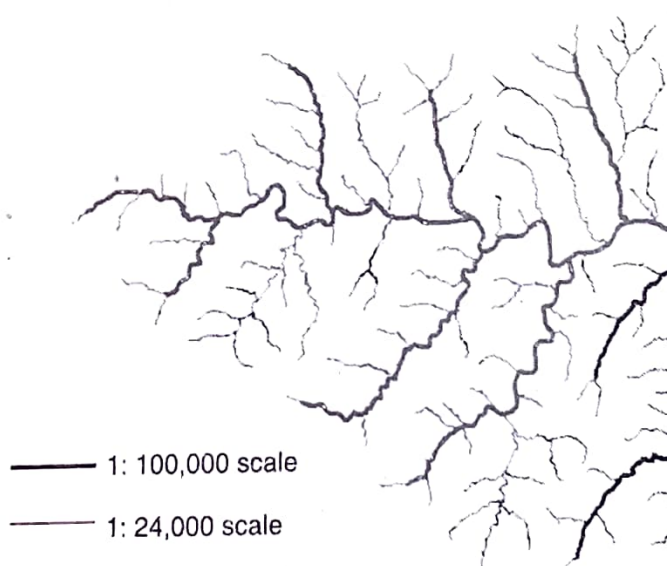


Fig. 1.6.1 : NHD at two spatial scales



- The **National Hydrography Dataset (NHD)** is the surface water component of the national map.

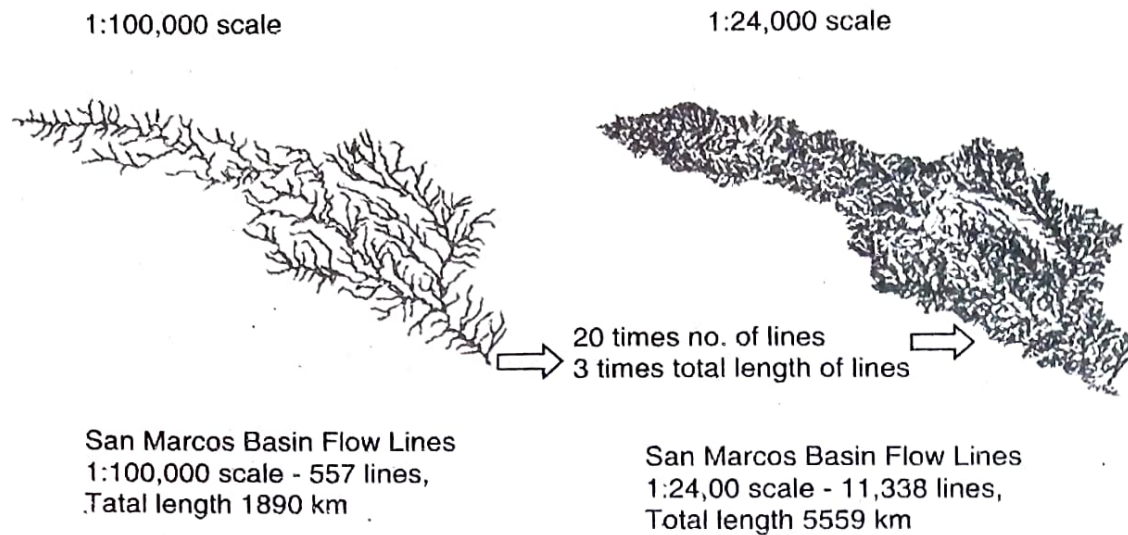


Fig. 1.6.2 : Streams and map scale

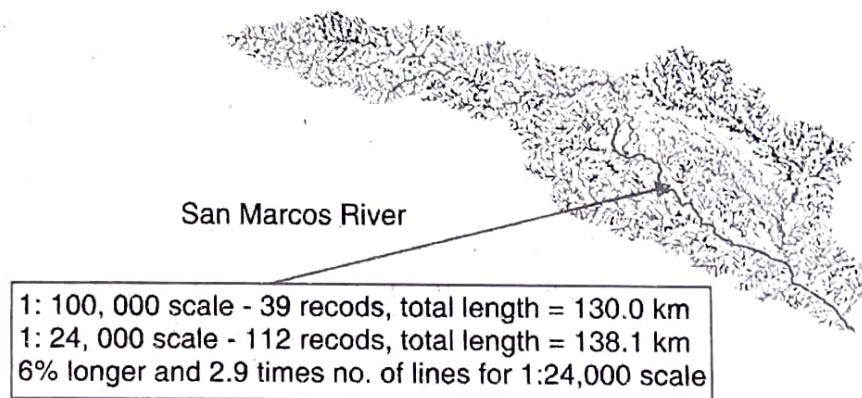


Fig. 1.6.3 : Length of the San Marcos river

- Accuracy is the degree to which information on a map or in a digital database matches true or accepted values.
- Precision refers to the level of measurement and exactness of description in a GIS database.

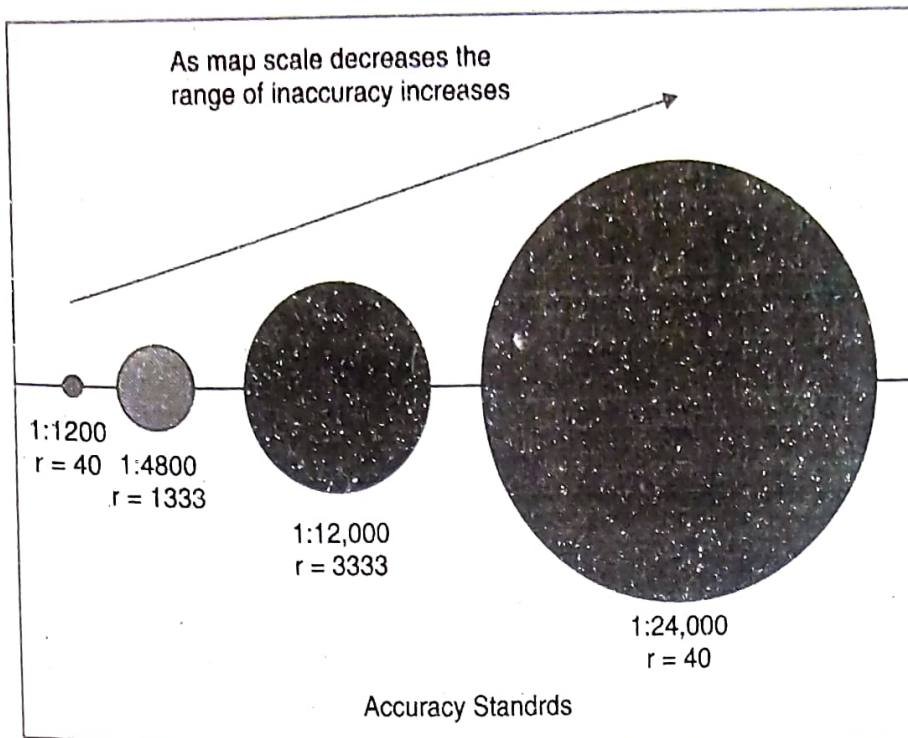


Fig. 1.6.4 : Relating scale and accuracy

- United States Geological Survey mapping standards : “Requirements for meeting horizontal accuracy as 90 percent of all measurable points must be within $1/30^{\text{th}}$ of an inch for maps at a scale of 1 : 20,000 or larger, and $1/50^{\text{th}}$ of an inch for maps at scales smaller than 1 : 20,000.”
- | | |
|--------------------------------|---------------------------------|
| ○ 1 : 1,200 $\pm$ 3.33 feet | ○ 1 : 2,400 $\pm$ 6.67 feet |
| ○ 1 : 4,800 $\pm$ 13.33 feet | ○ 1 : 10,000 $\pm$ 27.78 feet |
| ○ 1 : 12,000 $\pm$ 33.33 feet | ○ 1 : 1,24,000 $\pm$ 40.00 feet |
| ○ 1 : 63,360 $\pm$ 105.60 feet | ○ 1 : 100,000 $\pm$ 166.67 feet |

Syllabus Topic : Representation of Geographic Fields

1.7 Representation of Geographic Fields

Q. 1.7.1 Explain the representation of geographic fields. (Ref. Sec. 1.7)

(5 Marks)

It can be represented as discrete objects and continuous fields.

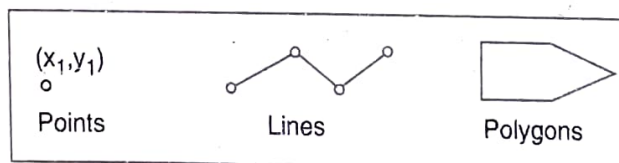


Fig. 1.7.1 : Discrete objects

(In the discrete object view, the world is empty, except where it is occupied by objects with well-defined boundaries that are instances of generally recognized categories.)

- Objects can be counted.
- Objects have dimensionality : 0-dimension (points), 1-dimension (lines), 2-dimensions (areas, polygons).
- 3-dimensions are problematic in GIS.
- The continuous field view represents the real world as a finite number of variables each one defined at every possible position.)
- Continuous fields, on the other hand, can be distinguished by what varies, and how smoothly.

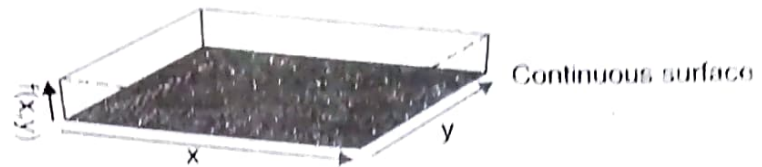


Fig. 1.7.2 : Continuous fields

Syllabus Topic : Representation of Geographic Objects

1.8 Representation of Geographic Objects

Q. 1.8.1 Explain the representation of geographic objects. (Ref Sec. 1.8) **(5 Marks)**

Geographic representation is defined as a representation of some part of the Earth's surface or near surface, at scales ranging from the architectural to the global short history of maps and the significance of their use historically like maps, any application of GIS requires attention to questions of what should be represented, and how shown in Fig. 1.8.1.

Environment Representation : The
Map Categories

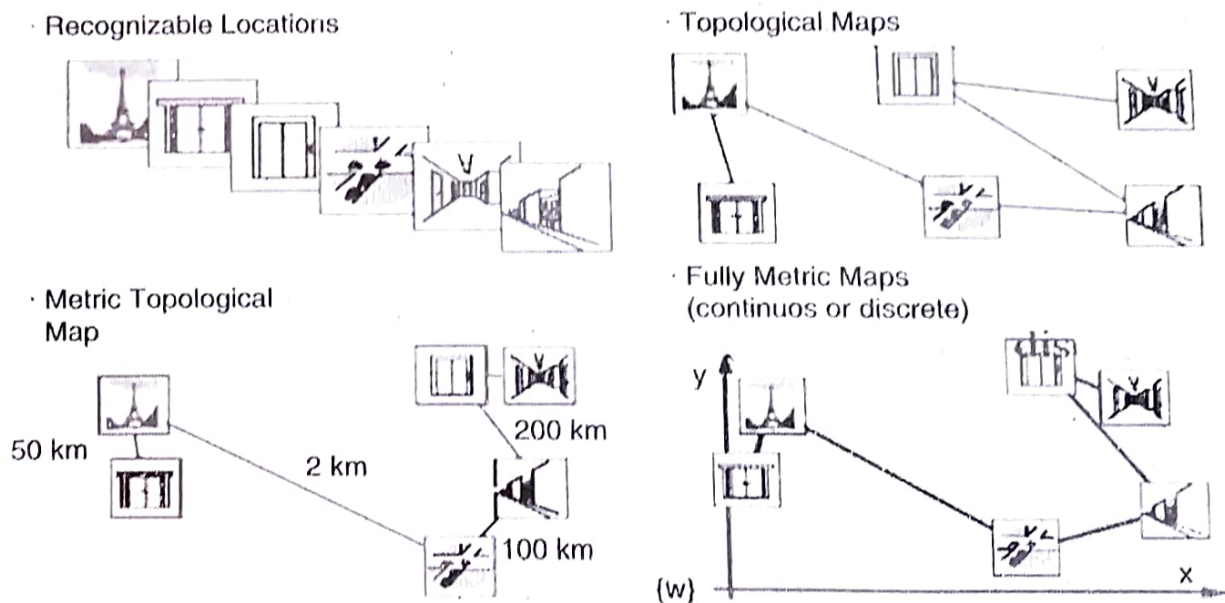


Fig. 1.8.1



In the scale-view, the block is looked as a common spatial object (polygon) in where it has spatial data and attributes data.

- In order to locate the spatial object in given OID (spatial Object ID) or BID (Block ID), it is necessary to add the "spatial object allocation table" to record the relation of ID and storage position of spatial object. In addition, for save information about child object or child block relation of block, the "block link table" is also necessary.

1.10 The Temporal Dimension

Q. 1.10.1 Write a short note on the temporal dimension. (Ref. Sec. 1.10)

(5 Marks)

- (With temporal data, you can analyze and visualize moving things (planes, satellites, storms), events (accidents or crimes), readings that vary over time but originate from stationary sensors (precipitation readings or traffic counts), or the change in the characteristic of a place over time (population change in a census tract or the change in sea surface temperature).)
- Although other cultures have different conceptions of time, we generally think of time as linear and unidirectional. Even events that are cyclic are conceived of as linear (last year, this year, next year) but repeating (winter, spring, summer, fall).

Temporal Data Types

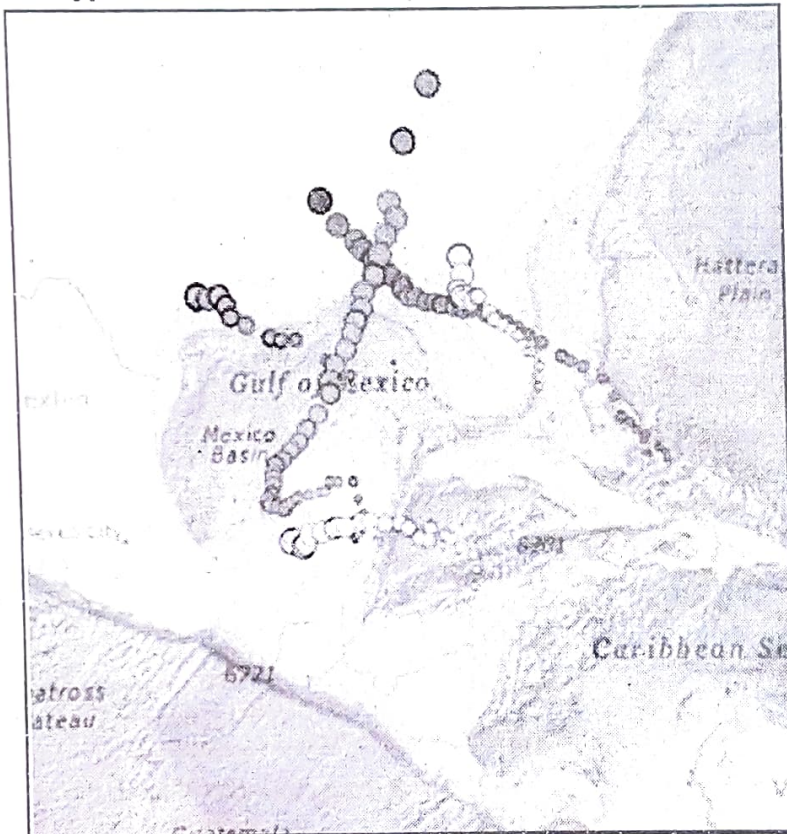


Fig. 1.10.1 : Moving Features